

Lesson 31B: Rectangular Equation and Polar Equation

Assessment Rubric: Rectangular & Polar Equations

Criteria	Exceptional (4)	Proficient (3)	Developing (2)	Beginner (1)
Conversion Accuracy (P → R)	Accurately converts polar equations to rectangular form, using all identities correctly with no errors.	Converts with minor algebraic errors that do not fundamentally change the curve type.	Correctly uses substitution, but fails to simplify the equation into a standard form.	Incorrect substitution of identities (e.g., uses $x = r \sin \theta$).
Conversion Accuracy (R → P)	Converts rectangular to polar equations, solving for r (when possible) and simplifying.	Correctly converts, but leaves the equation in implicit form rather than $r = f(\theta)$.	Substitution errors occur (e.g., misusing $r^2 = x^2 + y^2$).	Unable to apply core conversion formulas.
Graphing & Sketching	Accurately graphs the equation, identifying all key features (e.g., symmetry, radius, center, amplitude).	Sketches the graph correctly, but misses one minor feature or label (e.g., a specific point).	Sketches a roughly correct shape, but with wrong dimensions or wrong orientation.	Graph is incorrect or not provided.
Algebraic Manipulation	Employs efficient algebraic techniques (substitution, factoring, squaring) to reach final form.	Most algebraic steps are correct, but requires extra, unnecessary steps.	Several errors in factoring or expanding, making the final equation too complex.	Fails to simplify, leaving r and x/y mixed throughout.

**Interpretation
&
Understanding**

Confidently explains why a particular coordinate system is preferred for the given scenario.

Understands the relationship between the two forms but cannot explain the advantage of one over the other.

Can only perform the conversion if a direct formula is given.

No understanding of the relationship between rectangular and polar graphs.